

DW_fp_cmp

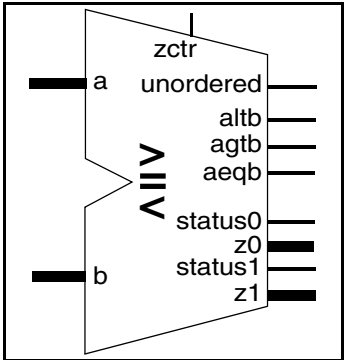
Floating-Point Comparator

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Features and Benefits

- The precision format is parameterizable for either IEEE single, double precision, or a user-defined custom format
- Exponents can range from 3 to 31 bits
- Significand and fractional part of the floating-point number can range from 2 to 256 bits
- Accuracy conforms to IEEE 754 Floating-point standard¹

Revision History



Description

DW_fp_cmp is a floating-point module that compares two floating-point numbers at inputs a and b. The format of these numbers is defined by the number of bits in the exponent (*exp_width*) and the number of bits in the significand (*sig_width*). Only one of the four outputs: agtb, altb, aeqb or unordered is set as a result of this operation. The input zctr is used to select the function that generates the z0 and z1 outputs. DW_fp_cmp also produces status0 and status1 with information about the z0 and z1 outputs. For detailed behavior, see [Table 1-5](#) on page 4 and [Table 1-6](#) on page 5.

[Table 1-1](#) contains pin descriptions and configuration parameters are listed in [Table 1-2](#) on page 2.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
a	<i>sig_width</i> + <i>exp_width</i> + 1 bits	Input	Floating-point number
b	<i>sig_width</i> + <i>exp_width</i> + 1 bits	Input	Floating-point number
altb	1 bit	Output	High when a is less than b
agtb	1 bit	Output	High when a is greater than b
aeqb	1 bit	Output	High when a is equal to b
unordered	1 bit	Output	High when one of the inputs is NaN and <i>ieee_compliance</i> = 1
z0	<i>sig_width</i> + <i>exp_width</i> + 1 bits	Output	■ When <i>zctr</i> is 0 or open, z0 = Min(a,b) ■ When <i>zctr</i> is 1, z0 = Max(a,b)
z1	<i>sig_width</i> + <i>exp_width</i> + 1 bits	Output	■ When <i>zctr</i> is 0 or open, z1 = Max(a,b) ■ When <i>zctr</i> is 1, z1 = Min(a,b)

1. For more information, see [IEEE 754 Compatibility](#) in the *Datapath Floating-Point Overview*.

Table 1-1 Pin Description (Continued)

Pin Name	Width	Direction	Function
status0 ^a	8 bits	Output	- status0[0:6]: Status flags corresponding to z0; for details, see STATUS Flags in the <i>Datapath Floating-Point Overview</i> - status0[7]^b: Set when operand a is passed to output z0
status1 ^a	8 bits	Output	- status1[0:6]: Status flags corresponding to z1; for details, see STATUS Flags in the <i>Datapath Floating-Point Overview</i> - status1[7]^b: Set when operand a is passed to output z1
zctr	1 bit	Input	Determines value passed to z0 and z1

a. DW_fp_cmp does not use bit 3 (the Tiny bit) or bit 4 (the Huge bit) of status0 and status1 because rounding is not performed

b. Bit 7 helps to trace the input that has the Max(a,b) or Min(a,b) value, based on the functionality of the z0 and z1 outputs (as determined by zctr).

For example, when zctr=0, z0=Min(a,b), z1=Max(a,b). So, when a=3 and b=4:

z0=Min(3,4)=3 and status0[7]=1

z1=Max(3,4)=4, and status1[7]=0

Looking at status0[7] and status1[7], you can see that the minimum input value (z0 output) is coming from input a (because status0[7] is set), and the maximum input value is coming from input b.

Table 1-2 Parameter Description

Parameter	Values	Description
sig_width	2 to 253 bits Default: 23	Word length of fraction field of floating-point numbers a, b, z0, and z1
exp_width	3 to 31 bits Default: 8	Word length of biased exponent of floating-point numbers a, b, z0, and z1
ieee_compliance	0 or 1 Default: 0	0: NaNs and denormals are unsupported; NaNs are considered infinities and denormals are considered zeros. 1: The generated architecture is fully compliant with IEEE 754 standard, including the use of denormals and NaNs.

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
rtl	Synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
DW02.DW_FP_CMP_CFG_SIM	Design unit name for VHDL simulation
dw/dw02/src/DW_fp_cmp_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_fp_cmp.v	Verilog simulation model source code

The parameter *ieee_compliance* controls the functionality of this component.

- When *ieee_compliance* = 0, the DW_fp_cmp component does not support NaNs and denormals. The *aeqb*, *altb*, and *agtb* outputs are generated considering denormal inputs as zeros, and the *z0* and *z1* outputs pass the values of inputs *a* and *b* without forcing them to zeros. Also, the *unordered* output is always zero in this configuration.
- When the *ieee_compliance* = 1, the component is fully compliant with the IEEE 754 standard and, therefore, operates with NaNs and denormals. The *unordered* output is set to one when a NaN value is applied as input.

For detailed behavior when *ieee_compliance* = 0, see [Table 1-5](#) on page 4.

For detailed behavior when *ieee_compliance* = 1, see [Table 1-6](#) on page 5.

Table 1-5 DW_fp_cmp Behavior When *sig_width* = 10, *exp_width* = 5, and *ieee_compliance* = 0

a	b	zctr	z0	z1	status0	status1	altb	agtb	aeqb	unordered
0_00011_0000110110 (normal number)	0_00000_0110000000 (denormal: zero)	0	0_00000_0110000000 (min(a,b)= b)	0_00011_0000110110 (max(a,b)= a)	00000001	10000000	0	1	0	0
0_00011_0000110110 (normal number)	0_00000_0110000000 (denormal: zero)	1	0_00011_0000110110 (max(a,b)= a)	0_00000_0110000000 (min(a,b)= b)	10000000	00000001	0	1	0	0
0_00000_0000110110 (denormal: zero)	1_00000_0110000000 (denormal: zero)	0	0_00000_0000110110 (a input)	1_00000_0110000000 (b input)	10000001	00000001	0	0	1	0
0_00000_0000110110 (denormal: zero)	1_00000_0110000000 (denormal: zero)	1	1_00000_0110000000 (b input)	0_00000_0000110110 (a input)	00000001	10000001	0	0	1	0
0_00000_0000000000 (zero)	1_00000_0000000000 (-zero)	0	0_00000_0000000000 (a input)	1_00000_0000000000 (b input)	10000001	00000001	0	0	1	0
0_00000_0000000000 (zero)	1_00000_0000000000 (-zero)	1	1_00000_0000000000 (b input)	0_00000_0000000000 (a input)	00000001	10000001	0	0	1	0
1_11111_0101100111 (NaN: -infinity)	0_10111_0010011010 (normal number)	0	1_11111_0101100111 (min(a,b)= a)	0_10111_0010011010 (max(a,b)= b)	10000010	00000000	1	0	0	0
1_11111_0101100111 (NaN: -infinity)	0_10111_0010011010 (normal number)	1	0_10111_0010011010 (max(a,b)= b)	1_11111_0101100111 (min(a,b)= a)	00000000	10000010	1	0	0	0
0_11111_0101100111 (NaN: +infinity)	1_11111_0010011010 (NaN: -infinity)	0	1_11111_0010011010 (min(a,b) = b)	0_11111_0101100111 (max(a,b) = a)	00000010	10000010	0	1	0	0
0_11111_0101100111 (NaN: +infinity)	1_11111_0010011010 (NaN: -infinity)	1	0_11111_0101100111 (max(a,b) = a)	1_11111_0010011010 (min(a,b) = b)	10000010	00000010	0	1	0	0
0_11111_0101100111 (NaN: +infinity)	0_11111_0010011010 (NaN: +infinity)	0	0_11111_0101100111 (a input)	0_11111_0010011010 (b input)	10000010	00000010	0	0	1	0
0_11111_0101100111 (NaN: +infinity)	0_11111_0010011010 (NaN: +infinity)	1	0_11111_0010011010 (b input)	0_11111_0101100111 (a input)	00000010	10000010	0	0	1	0

Table 1-6 DW_fp_cmp Behavior When *sig_width* = 10, *exp_width* = 5, and *ieee_compliance* = 1

a	b	zctr	z0	z1	status0	status1	altb	agtb	aeqb	unordered
0_00011_0000110110 (normal number)	0_00000_0110000000 (denormal)	0	0_00000_0110000000 (min(a,b)= b)	0_00011_0000110110 (max(a,b)= a)	00000000	10000000	0	1	0	0
0_00011_0000110110 (normal number)	0_00000_0110000000 (denormal)	1	0_00011_0000110110 (max(a,b)= a)	0_00000_0110000000 (min(a,b)= b)	10000000	00000000	0	1	0	0
0_00000_0000110110 (denormal)	1_00000_0110000000 (denormal)	0	1_00000_0110000000 (min(a,b)= b)	0_00000_0000110110 (max(a,b)= a)	00000000	10000000	0	1	0	0
0_00000_0000110110 (denormal)	1_00000_0110000000 (denormal)	1	0_00000_0000110110 (max(a,b)= a)	1_00000_0110000000 (min(a,b)= b)	10000000	00000000	0	1	0	0
0_00000_0000000000 (zero)	1_00000_0000000000 (-zero)	0	0_00000_0000000000 (a input)	1_00000_0000000000 (b input)	10000001	00000001	0	0	1	0
0_00000_0000000000 (zero)	1_00000_0000000000 (-zero)	1	1_00000_0000000000 (b input)	0_00000_0000000000 (a input)	00000001	10000001	0	0	1	0
1_11111_0101100111 (NaN)	0_10111_0010011010 (normal number)	0	1_11111_0101100111 (a input)	0_10111_0010011010 (b input)	10000100	00000000	0	0	0	1
1_11111_0101100111 (NaN)	0_10111_0010011010 (normal number)	1	1_11111_0101100111 (a input)	0_10111_0010011010 (b input)	10000100	00000000	0	0	0	1
0_11111_0101100111 (NaN)	1_11111_0010011010 (NaN)	0	0_11111_0101100111 (a input)	1_11111_0010011010 (b input)	10000100	00000100	0	0	0	1
0_11111_0101100111 (NaN)	1_11111_0010011010 (NaN)	1	0_11111_0101100111 (a input)	1_11111_0010011010 (b input)	10000100	00000100	0	0	0	1
0_11111_0101100111 (NaN)	0_11111_0010011010 (NaN)	0	0_11111_0101100111 (a input)	0_11111_0010011010 (b input)	10000100	00000100	0	0	0	1
0_11111_0101100111 (NaN)	0_11111_0010011010 (NaN)	1	0_11111_0101100111 (a input)	0_11111_0010011010 (b input)	10000100	00000100	0	0	0	1

Related Topics

- [Datapath – Floating-Point Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```

library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_Foundation_comp.all;
-- If using numeric types from std_logic_arith package,
-- comment the preceding line and uncomment the following line:
-- use DWARE.DW_Foundation_comp_arith.all;

entity DW_fp_cmp_inst is
    generic (
        inst_sig_width : POSITIVE := 23;
        inst_exp_width  : POSITIVE := 8;
        inst_ieee_compliance : INTEGER := 0
    );
    port (
        inst_a : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
        inst_b : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
        inst_zctr : in std_logic;
        aeqb_inst : out std_logic;
        altb_inst : out std_logic;
        agtb_inst : out std_logic;
        unordered_inst : out std_logic;
        z0_inst : out std_logic_vector(inst_sig_width+inst_exp_width downto 0);
        z1_inst : out std_logic_vector(inst_sig_width+inst_exp_width downto 0);
        status0_inst : out std_logic_vector(7 downto 0);
        status1_inst : out std_logic_vector(7 downto 0)
    );
end DW_fp_cmp_inst;

architecture inst of DW_fp_cmp_inst is

begin
    -- Instance of DW_fp_cmp
    U1 : DW_fp_cmp
        generic map ( sig_width => inst_sig_width, exp_width => inst_exp_width,
            ieee_compliance => inst_ieee_compliance )
        port map ( a => inst_a, b => inst_b, zctr => inst_zctr, aeqb => aeqb_inst,
            altb => altb_inst, agtb => agtb_inst, unordered => unordered_inst, z0 =>
            z0_inst, z1 => z1_inst, status0 => status0_inst, status1 => status1_inst );
end inst;

```

```
-- pragma translate_off
configuration DW_fp_cmp_inst_cfg_inst of DW_fp_cmp_inst is
  for inst
    end for; -- inst
end DW_fp_cmp_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_fp_cmp_inst( inst_a, inst_b, inst_zctr, aeqb_inst, altb_inst,
                      agtb_inst, unordered_inst, z0_inst, z1_inst, status0_inst,
                      status1_inst );

parameter sig_width = 23;
parameter exp_width = 8;
parameter ieee_compliance = 0;

input [sig_width+exp_width : 0] inst_a;
input [sig_width+exp_width : 0] inst_b;
input inst_zctr;
output aeqb_inst;
output altb_inst;
output agtb_inst;
output unordered_inst;
output [sig_width+exp_width : 0] z0_inst;
output [sig_width+exp_width : 0] z1_inst;
output [7 : 0] status0_inst;
output [7 : 0] status1_inst;

// Instance of DW_fp_cmp
DW_fp_cmp #(sig_width, exp_width, ieee_compliance)
  U1 ( .a(inst_a), .b(inst_b), .zctr(inst_zctr), .aeqb(aeqb_inst),
      .altb(altb_inst), .agtb(agtb_inst), .unordered(unordered_inst),
      .z0(z0_inst), .z1(z1_inst), .status0(status0_inst),
      .status1(status1_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	Updated date and version; no technical changes
December 2023	DWBB_202312.0	Updated date and version; no technical changes
October 2021	DWBB_202106.3	- Added footnote about bits 3 and 4 of <code>status0</code> and <code>status1</code> in Table 1-1 on page 1 - Clarified the releases noted in this table
July 2021	DWBB_202106.1	- Clarified behavior when <code>ieee_compliance</code> = 0 in Table 1-2 on page 2 and in text on page 3 - Clarified the purpose of <code>status0[7]</code> and <code>status1[7]</code> on page 2 - Added detailed behavior in Table 1-5 on page 4 and Table 1-6 on page 5
May 2020	DWBB_201912.5	Corrected the example in “HDL Usage Through Component Instantiation - VHDL” on page 6
April 2020	DWBB_201912.3	- Corrected some formatting in this datasheet - Added this Revision History table and document links

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